

Hume City Council CyberSafe Services

Online Scam Detection and Reporting System

Overview

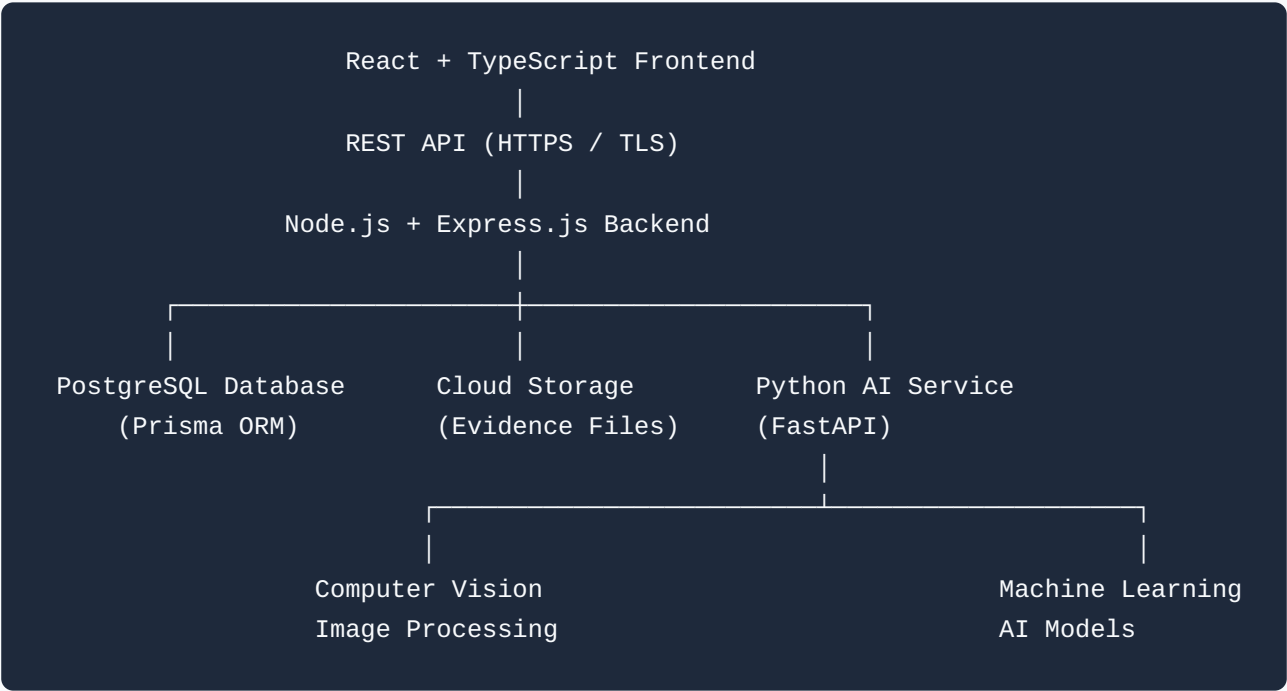
The **Online Scam Detection and Reporting System** is designed as a modern, cloud-ready, AI-enabled cybersecurity SaaS platform. The technology stack has been carefully selected to meet the project's functional and non-functional requirements while remaining **free, open-source, reliable, secure, scalable, and future-proof**.

The architecture follows a **modular microservice-inspired approach**, allowing future integration of Artificial Intelligence (AI), Computer Vision, Machine Learning, Optical Character Recognition (OCR), and advanced cybersecurity services without requiring major architectural changes.

The selected technologies emphasize:

- High performance
- Enterprise-grade security
- Scalability
- Maintainability
- Developer productivity
- AI readiness
- Cross-platform compatibility

Architecture Overview



Technology Stack

Layer	Technology	Purpose
Frontend	React.js	User Interface
Language	TypeScript	Type-safe development
Build Tool	Vite	Fast development and builds
Styling	Tailwind CSS	Responsive UI
Component Library	shadcn/ui	Accessible UI components
Icons	Lucide React	Modern icon library
State Management	React Context API	Global application state
Forms	React Hook Form	Form handling
Validation	Zod	Input validation

Layer	Technology	Purpose
Backend	Node.js	Runtime environment
Framework	Express.js	REST API development
API Style	REST API	Client-server communication
Database	PostgreSQL	Relational database
ORM	Prisma ORM	Database management
Authentication	JWT	Secure authentication
Password Encryption	bcrypt	Password hashing
File Storage	Cloudinary	Evidence storage
Maps	Leaflet + OpenStreetMap	Scam location visualization
Charts	Recharts	Analytics dashboard
Email	Nodemailer	Email notifications
Logging	Winston	Application logging
Security	Helmet	HTTP security headers
Rate Limiting	Express Rate Limit	API protection
Deployment	Docker	Containerization
Version Control	Git + GitHub	Source control
AI Service	FastAPI (Python)	AI Microservice
Computer Vision	OpenCV	Image processing
OCR	EasyOCR	Text extraction
Deep Learning	PyTorch	Machine learning
AI Framework	TensorFlow	Deep learning

Layer	Technology	Purpose
Environment	Docker Compose	Local development

Frontend Technologies

| React.js

React is selected because it is the industry-standard library for developing interactive web applications.

Benefits

- Component-based architecture
- High performance through Virtual DOM
- Large ecosystem
- Easy maintenance
- Scalable application structure
- Excellent community support

React is ideal for: Dashboard, User Profile, Report Management, Analytics, Administration Panel, Interactive Maps, and Notifications.

| TypeScript

TypeScript extends JavaScript with static typing.

Benefits

- Early error detection
- Improved code quality
- Better IntelliSense
- Easier maintenance
- Improved scalability
- Better developer productivity

| Vite

Vite is selected as the frontend build tool.

Benefits

- Extremely fast development server
- Lightning-fast Hot Module Replacement
- Optimized production builds
- Modern ES Modules

| Tailwind CSS

Tailwind CSS provides utility-first styling.

Benefits

- Responsive design
- Clean code
- Highly customizable
- Minimal CSS maintenance
- Excellent performance

| shadcn/ui

Provides professional, accessible UI components. Benefits include:

- WCAG accessibility
- Modern design
- Reusable components
- Dark mode support
- Minimal dependencies

Backend Technologies

| Node.js

Node.js provides a high-performance JavaScript runtime suitable for scalable web applications.

Benefits

- Non-blocking architecture
- Event-driven

- Fast API development
- Huge package ecosystem
- Excellent API integration

| Express.js

Express.js is used to develop RESTful APIs.

Benefits

- Lightweight & Flexible
- Fast & Easy routing
- Middleware support
- Secure architecture

| REST API

REST APIs provide communication between frontend and backend. Advantages include:

- Stateless communication
- Easy integration
- Platform independent
- Mobile friendly
- AI service friendly

Database

| PostgreSQL

PostgreSQL is selected instead of MySQL because it provides superior enterprise features.

Advantages

- ACID compliance
- Advanced indexing & JSON support
- Strong security & Better concurrency
- Excellent scalability & Full SQL compliance

PostgreSQL is ideal for: User management, Scam reports, Evidence metadata, Audit logs, Notifications, Analytics, and Risk scores.

| Prisma ORM

Prisma simplifies database interaction.

Benefits

- Type-safe queries
- Migration management
- Auto-generated models
- Easy maintenance & Reduced SQL complexity

Authentication & Security

| JWT Authentication

Authentication is implemented using JSON Web Tokens. Features:

- Secure login
- Stateless authentication
- Refresh Tokens
- Session management

| bcrypt

Passwords are never stored in plaintext. bcrypt provides:

- Secure hashing
- Salt generation
- Resistance against brute-force attacks

| Helmet

Helmet secures HTTP headers. Protection includes:

- Clickjacking
- MIME sniffing

- XSS mitigation
- Content Security Policy

| Express Rate Limit

Protects against: API abuse, Brute-force attacks, and DDoS attempts.

| HTTPS

All communication uses: SSL/TLS Encryption, Secure API communication, Encrypted authentication tokens.

File Storage & Maps

| Cloudinary

Cloudinary is selected for evidence storage (Images, Screenshots, Documents, Evidence files).

Advantages

- Free tier & CDN delivery
- Image optimization & Secure uploads
- Easy API integration

The database stores only file metadata and secure URLs.

| Leaflet & OpenStreetMap

Interactive mapping library used for Scam hotspots, Community reports, and Regional alerts. OpenStreetMap provides free map tiles with no licensing fees and is community maintained.

Additional Services

| Charts & Analytics: Recharts

Used for: Risk score visualization, Scam trends, Dashboard analytics, Report statistics, and User activity graphs.

Email Service: Nodemailer

Provides: Email verification, Password reset, Notification emails, and Report status updates.

Logging: Winston

Logs: Errors, User activity, Security events, System events, and Audit logs.

Version Control: Git & GitHub

Tracks source code, branches, and releases. Used for Collaboration, Pull Requests, CI/CD, Version history, and Documentation.

Containerization: Docker

Docker packages the application into portable containers. Benefits:

- Consistent environments
- Easy deployment
- Scalability & Isolation
- Cloud readiness

AI & Computer Vision Architecture

To ensure future scalability, Artificial Intelligence components are separated into a dedicated Python microservice.

```
graph TD
    A[React Frontend] --> B[Express API]
    B --> C[FastAPI AI Service]
    C --- D[OpenCV]
    C --- E[EasyOCR]
    C --- F[PyTorch]
    C --- G[TensorFlow]
    C --- H[AI Models]
```

This architecture allows AI models to evolve independently from the main application.

Future AI Technologies & Features

Technologies

- **FastAPI:** Acts as the AI service. High performance, Async support, Easy API development.
- **OpenCV:** Screenshot analysis, Image enhancement, Logo/QR detection, Duplicate detection.
- **EasyOCR:** Extracts text from scam screenshots, emails, SMS, fake invoices, payment receipts.
- **PyTorch & TensorFlow:** Deep learning, NLP, Image classification, Scam prediction models, Object detection.

Future Features

- Scam text classification & Email phishing detection
- URL risk prediction & Image forgery detection
- Duplicate report detection & AI-powered chatbot
- Cybersecurity recommendation engine & Threat intelligence integration
- Risk prediction models & Natural Language Processing (NLP)

External API Integration & Deployment Stack

External APIs

The system architecture supports integration with: VirusTotal API, Google Safe Browsing API, AbuseIPDB API, WHOIS Lookup API, OpenAI API, Google Gemini API, Anthropic Claude API, Local LLMs (Ollama), Government cybersecurity services, and Email reputation services.

Deployment Stack

Component	Technology
Frontend Hosting	Vercel
Backend Hosting	Render

Component	Technology
Database	Neon PostgreSQL
File Storage	Cloudinary
Source Control	GitHub
Containerization	Docker
Monitoring	UptimeRobot

Why This Technology Stack?

The selected technology stack satisfies all project requirements by providing:

- Completely free and open-source core technologies
- Enterprise-grade security and excellent scalability
- Cloud-native architecture and high performance
- AI-ready design with future Machine Learning integration
- Modern development workflow with strong community support

This architecture enables the Online Scam Detection and Reporting System to evolve from a traditional web application into an intelligent cybersecurity platform capable of leveraging Artificial Intelligence, Computer Vision, Image Processing, and Machine Learning technologies while maintaining security, reliability, and scalability.